

Two-Point Oxygen Sensor Calibration

Mathematical derivation of a and b , SCP scaling, and a complete RSLogix 500 implementation

Based on the supplied Project 7 specification: 0–40% O₂ sensor, 0% and 30% calibration gases, 30-second sampling intervals.

Core idea

The calibration assumes a linear sensor model $Y = aX + b$. Two known gas concentrations provide two points, allowing both the gain a and offset b to be estimated. The PLC then represents the same calibrated line through the SCP instruction by updating its **Input Min** and **Input Max** parameters.

1 Specification and Address Map

The process has two operating modes: normal sampling and calibration. During calibration, the PLC first exposes the sensor to 0% O₂ for 30 seconds, then to 30% O₂ for 30 seconds. Average raw readings from the two periods are used to update the scale.

Address	Purpose
N7:0	Raw oxygen-sensor input signal
B3:0/0	Calibrate pushbutton
0:0/0	0% calibration-gas valve, energized open
0:0/1	30% calibration-gas valve, energized open
N7:1	Measured oxygen concentration, percent
N7:2	SCP Input Min; default 0
N7:3	SCP Input Max; default 16383

At the default scale, $N7:0 = 8192$ should produce approximately 20% O₂ because 8192 is approximately half of the 0–16383 raw range.

2 Mathematics of Two-Point Calibration

2.1 Linear model

Let

$$Y = aX + b, \quad (1)$$

where X is the raw PLC input and Y is oxygen concentration in percent. The two unknown parameters have direct physical meanings:

- a is the **gain or slope**: how many percentage points of oxygen correspond to one raw count.
- b is the **offset or intercept**: the correction required because a real sensor may not produce raw zero at 0% O₂.

Let X_0 be the average raw reading under 0% gas and X_{30} the average raw reading under 30% gas. The two known calibration points are

$$(X_0, 0), \quad (X_{30}, 30).$$

2.2 Estimating the slope a

The slope through two points is

$$a = \frac{Y_2 - Y_1}{X_2 - X_1}.$$

Therefore,

$$\boxed{a = \frac{30}{X_{30} - X_0}}. \quad (2)$$

The denominator $X_{30} - X_0$ is the sensor's measured raw span for a known 30-percentage-point change in oxygen concentration.

2.3 Estimating the intercept b

Use the 0% calibration point in $Y = aX + b$:

$$0 = aX_0 + b.$$

Hence

$$\boxed{b = -aX_0} \quad (3)$$

and, after substituting the expression for a ,

$$\boxed{b = -\frac{30X_0}{X_{30} - X_0}}. \quad (4)$$

Thus the calibrated line can be written in either of two equivalent forms:

$$Y = \frac{30}{X_{30} - X_0}X - \frac{30X_0}{X_{30} - X_0}, \quad (5)$$

$$Y = 30 \frac{X - X_0}{X_{30} - X_0}. \quad (6)$$

$$\boxed{Y = 30 \frac{X - X_0}{X_{30} - X_0}} \quad (7)$$

Why b does not need a separate measurement

Once the slope a has been found from the two calibration points, the measured zero-gas point forces the calibrated line to pass through $(X_0, 0)$. That condition uniquely determines $b = -aX_0$.

2.4 Worked example from the project test

For the aged-sensor test,

$$X_0 = 100, \quad X_{30} = 11000.$$

Then

$$a = \frac{30}{11000 - 100} = \frac{30}{10900} \approx 0.00275229,$$

$$b = -a(100) \approx -0.275229.$$

Therefore

$$\boxed{Y \approx 0.00275229X - 0.275229}. \quad (8)$$

The two known gases verify the result exactly apart from numerical rounding:

$$X = 100 \Rightarrow Y \approx 0\%, \quad X = 11000 \Rightarrow Y \approx 30\%.$$

3 How the Same Line Is Represented by SCP

The project does not ask the PLC to store a and b directly. Instead, **SCP** maps a raw input interval to the fixed engineering interval 0–40% O₂.

The zero point is direct:

$$\boxed{\text{Input Min} = X_0.} \quad (9)$$

The second certified gas is 30%, while the desired full engineering range ends at 40%. Therefore the PLC linearly extrapolates the measured 0–30% raw span to the raw value expected at 40%:

$$\boxed{\text{Input Max} = X_0 + \frac{40}{30}(X_{30} - X_0).} \quad (10)$$

For $X_0 = 100$ and $X_{30} = 11000$,

$$\text{Input Max} = 100 + \frac{40}{30}(10900) \approx 14633.33.$$

The values stored by the PLC are therefore approximately

$$\boxed{N7:2 = 100, \quad N7:3 = 14633.}$$

The **SCP** equation becomes

$$Y = 40 \frac{X - N7:2}{N7:3 - N7:2}. \quad (11)$$

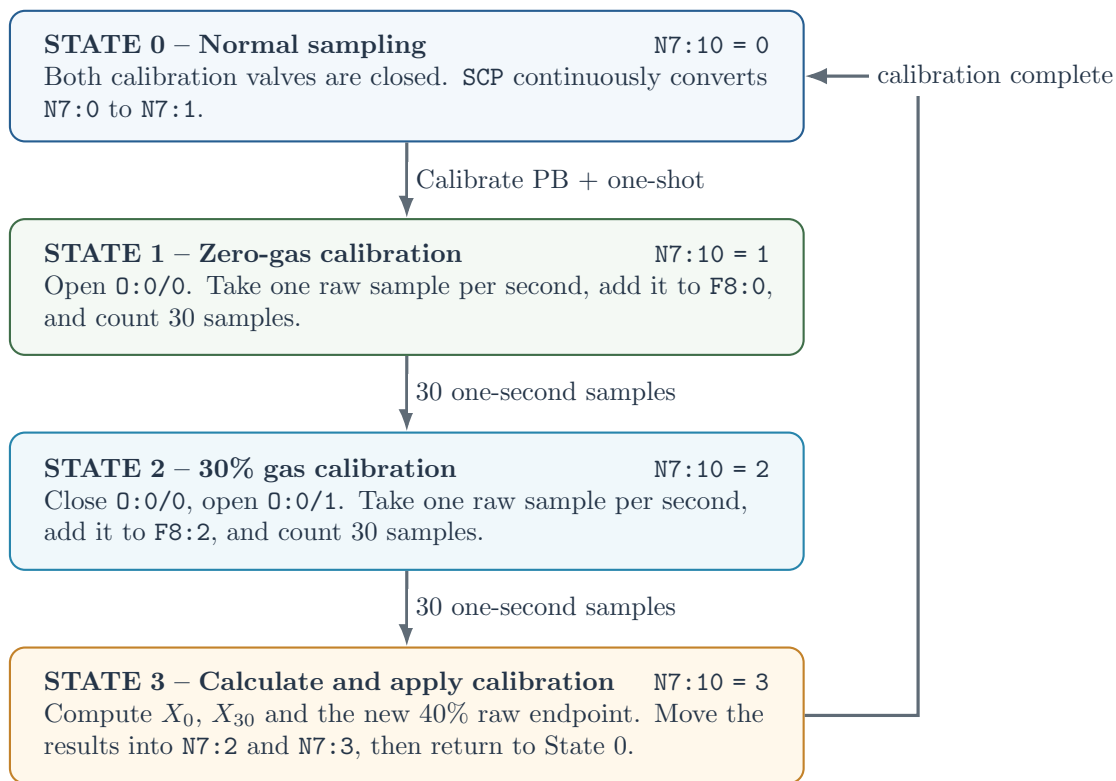
Substituting the expression for **Input Max** reduces this equation back to

$$Y = 30 \frac{X - X_0}{X_{30} - X_0},$$

proving that the explicit $aX + b$ model and the **SCP** implementation are the same straight line.

4 Calibration Sequence

The PLC implementation below uses a four-state sequence. A vertical layout is used intentionally so that the operating logic can be read top-to-bottom without crossing arrows.



5 PLC Working Memory

The following additional addresses make the state machine and arithmetic explicit.

Address	Purpose
N7:10	Calibration state: 0 = normal, 1 = zero gas, 2 = 30% gas, 3 = calculate
F8:0	Sum of the 30 zero-gas samples
F8:1	Zero-gas average X_0
F8:2	Sum of the 30 test-gas samples
F8:3	30% gas average X_{30}
F8:4	Temporary span, $X_{30} - X_0$
F8:5	Temporary full-scale calculation / computed Input Max
T4:0	Repeating one-second sample timer
T4:1	Repeating one-second sample timer
C5:0	Sample counter, preset 30
C5:1	Sample counter, preset 30
B3:0/2	One-shot storage for calibration pushbutton
B3:0/3	One-shot storage for State 1 sample pulse
B3:0/4	One-shot storage for State 2 sample pulse

Why floating-point sums are used

Thirty raw readings can exceed the signed 16-bit integer range. For example, $30 \times 16383 = 491,490$. Accumulating into F8 files avoids integer overflow and also preserves fractional averages.

6 Complete RSLogix 500 Program, Rung by Rung

The program below follows the Project 7 sequence and does not use CPT. The arithmetic is decomposed into SUB, MUL, DIV, ADD, and MOV instructions.

Rung 000 – First-scan initialization

Initialize the default scale and state only when the processor first enters Run mode.

```
XIC S:1/15 MOV 0 N7:2
XIC S:1/15 MOV 16383 N7:3
XIC S:1/15 MOV 0 N7:10
```

Rung 001 – Start a calibration cycle

Only start from State 0. The one-shot prevents a held calibration button from restarting the sequence every scan. Clear both sample sums and reset the shared timer/counter before entering State 1.

```
EQU N7:10 0 XIC B3:0/0 ONS B3:0/2
MOV 0.0 F8:0
MOV 0.0 F8:2
RES T4:0
RES C5:0
MOV 1 N7:10
```

Rung 002 – Zero-gas valve

The valve is energized only while the state equals 1.

```
EQU N7:10 1 OTE 0:0/0
```

Rung 003 – One-second timer for State 1

Use a 1.00-second preset. For a 0.01-second time base, use PRE = 100; for a 1.0-second time base, use PRE = 1.

```
EQU N7:10 1 TON T4:0 PRE 100 ACC 0
```

Rung 004 – Acquire one zero-gas sample

When the timer reaches done, add the current raw input to the zero-gas sum, increment the sample counter once, and reset the timer so that the next one-second interval begins.

```
EQU N7:10 1 XIC T4:0/DN ONS B3:0/3
ADD F8:0 N7:0 F8:0
CTU C5:0 PRE 30 ACC 0
RES T4:0
```

Rung 005 – Finish State 1 and compute X_0

When 30 samples have been counted, compute the average, reset the shared timer/counter, and advance to State 2.

```
EQU N7:10 1 XIC C5:0/DN
DIV F8:0 30.0 F8:1
RES C5:0
RES T4:0
MOV 2 N7:10
```

Rung 006 – 30% calibration-gas valve

Entering State 2 automatically de-energizes 0:0/0 and energizes 0:0/1.

```
EQU N7:10 2 OTE 0:0/1
```

Rung 007 – One-second timer for State 2

We do not want to reuse the same timer T4:0 for the second sampling period. Doing so breaks the program.

```
EQU N7:10 2 TON T4:1 PRE 100 ACC 0
```

Rung 008 – Acquire one 30% gas sample

A separate one-shot storage bit is used for this sampling rung.

```
EQU N7:10 2 XIC T4:1/DN ONS B3:0/4
ADD F8:2 N7:0 F8:2
CTU C5:1 PRE 30 ACC 0
RES T4:1
```

Rung 009 – Finish State 2 and compute X_{30}

After 30 samples, compute the second average and enter the calculation state.

```
EQU N7:10 2 XIC C5:1/DN
DIV F8:2 30.0 F8:3
RES C5:1
RES T4:1
MOV 3 N7:10
```

Rung 010 – Compute measured raw span

```
EQU N7:10 3 SUB F8:3 F8:1 F8:4
; F8:4 = X30 - X0
```

Rung 011 – Scale the 30% span toward 40%

Multiply before dividing. This avoids any risk that an integer evaluation of $40/30$ would truncate to 1.

```
EQU N7:10 3 MUL F8:4 40.0 F8:5
; F8:5 = (X30 - X0) * 40
```

Rung 012 – Divide by the 30% reference concentration

```
EQU N7:10 3 DIV F8:5 30.0 F8:5
; F8:5 = (X30 - X0) * 40 / 30
```

Rung 013 – Restore the zero offset

```
EQU N7:10 3 ADD F8:5 F8:1 F8:5
; F8:5 = X0 + (40/30)*(X30 - X0)
```

Rung 014 – Apply new SCP Input Min

```
EQU N7:10 3 MOV F8:1 N7:2
; N7:2 = X0
```

Rung 015 – Apply new SCP Input Max

Moving from a floating-point source to an integer destination converts the computed endpoint to an integer suitable for SCP.

```
EQU N7:10 3 MOV F8:5 N7:3
; N7:3 = predicted raw value at 40% O2
```

Rung 016 – End calibration

This rung must appear *after* Rungs 010–015. RSLogix scans top-to-bottom; resetting the state earlier would prevent the later State 3 calculations from executing on that scan.

```
EQU N7:10 3 MOV 0 N7:10
```

Rung 017 – Continuous oxygen measurement

The SCP rung can execute continuously. During normal operation it uses the most recently calibrated limits.

```
SCP
Source: N7:0
Input Min: N7:2
Input Max: N7:3
Scaled Min: 0
Scaled Max: 40
Destination: N7:1
```

RSLogix entry note

If your processor/editor rejects mixed integer/float operands in an arithmetic instruction such as `ADD F8:0 N7:0 F8:0`, first `MOV N7:0` into a temporary float (for example `F8:6`) and then add `F8:6` to the accumulator. The calibration mathematics is unchanged.

7 Program Operation by Scan

The logic is intentionally ordered so that state transitions are deterministic:

1. A calibration-button one-shot changes `N7:10` from 0 to 1.
2. While State 1 is active, only the zero-gas valve is energized. Every timer completion contributes one sample to `F8:0` and one count to `C5:0`.
3. On the 30th sample, Rung 005 computes X_0 and changes the state to 2. On the same scan, lower rungs now see State 2, so the 30% valve becomes active.
4. The second 30-sample period accumulates into `F8:2`. On completion, the state becomes 3.
5. Rungs 010–015 execute the full calibration arithmetic in sequence during State 3.
6. Rung 016 returns the state to 0 only after all new scale values have been written.
7. The continuous SCP rung then interprets `N7:0` using the updated `N7:2` and `N7:3`.

8 Verification Against the Project Tests

8.1 Default scale

With

$$N7:2 = 0, \quad N7:3 = 16383,$$

setting `N7:0 = 8192` should give approximately

$$N7:1 \approx 20\%.$$

8.2 Calibration with ideal zero and 30% raw value 12288

If the zero-gas average is $X_0 = 0$ and the 30% average is $X_{30} = 12288$,

$$N7:3 = 0 + \frac{40}{30}(12288) = 16384 \text{ (approximately).}$$

The calibrated scale therefore produces approximately 0%, 30%, and 40% at raw values 0, 12288, and 16383/16384 respectively.

8.3 Calibration with offset and gain error

For the harder test,

$$X_0 = 100, \quad X_{30} = 11000.$$

The PLC calculates

$$\begin{aligned} \text{span} &= 11000 - 100 = 10900, \\ 40\% \text{ span} &= 10900 \cdot \frac{40}{30} = 14533.33, \\ \text{Input Max} &= 14533.33 + 100 = 14633.33. \end{aligned}$$

Thus

$$N7:2 \approx 100, \quad N7:3 \approx 14633.$$

Expected checks are then

$$100 \rightarrow 0\%, \quad 11000 \rightarrow 30\%, \quad 14633 \rightarrow 40\%.$$

9 Engineering Interpretation

The two gases identify two independent error mechanisms:

- **Offset error** is measured by X_0 . A nonzero raw reading at 0% changes the intercept b and becomes the new SCP Input Min.
- **Gain/span error** is measured by $X_{30} - X_0$. If sensor sensitivity changes with age, the raw difference associated with a 30% oxygen change also changes; this alters the slope a and therefore the computed 40% raw endpoint.

The calibration does not physically repair the sensor. It updates the PLC's interpretation of the raw signal so that the two certified reference gases map to their known concentrations.

10 Compact Formula Reference

Linear model: $Y = aX + b$

Slope: $a = \frac{30}{X_{30} - X_0}$

Intercept: $b = -aX_0$

Calibrated equation: $Y = 30 \frac{X - X_0}{X_{30} - X_0}$

SCP Input Min: $N7:2 = X_0$

SCP Input Max: $N7:3 = X_0 + \frac{40}{30}(X_{30} - X_0)$